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**ARTIFICIAL INTELLIGENCE (MACHINE LEARNING & DEEP
LEARNING)**

NAVTTTC PMYSDP

LECTURE No 05

SIR SUHAIL IMRAN

✓ For Loop

```
# Range Function (start, stop, step)
```

```
for i in range(0, 3, 1):  
    print("Python", end = "\t")
```

```
Python Python Python
```

```
for i in range(0, 3):  
    print(f"{i} Python")
```

```
0 Python  
1 Python  
2 Python
```

```
for i in range(0, 3):  
    print(i*2)
```

```
0  
2  
4
```

```
for i in range(0,3):  
    print(2, i, 2*i)
```

```
2 0 0  
2 1 2  
2 2 4
```

```
Number = int(input("Enter a number: "))  
for i in range(1,11,1):  
    print(f"{Number} x {i} = {Number * i}")
```

```
Enter a number: 5  
5 x 1 = 5  
5 x 2 = 10  
5 x 3 = 15  
5 x 4 = 20  
5 x 5 = 25  
5 x 6 = 30  
5 x 7 = 35  
5 x 8 = 40  
5 x 9 = 45  
5 x 10 = 50
```

```
Number = int(input("Enter a number: "))  
min = 1  
max = 11  
for i in range(min,max,1):  
    print(f"{Number} x {max - i} = {Number * (max - i)}")
```

```
Enter a number: 2  
2 x 10 = 20  
2 x 9 = 18  
2 x 8 = 16  
2 x 7 = 14  
2 x 6 = 12  
2 x 5 = 10  
2 x 4 = 8  
2 x 3 = 6  
2 x 2 = 4  
2 x 1 = 2
```

```
Number = int(input("Enter a number: "))  
min = 0  
max = 10  
for i in range(max,min, -1):  
    print(f"{Number} x {i} = {Number * (i)}")
```

```
Enter a number: 2  
2 x 10 = 20  
2 x 9 = 18  
2 x 8 = 16  
2 x 7 = 14  
2 x 6 = 12
```

```
2 x 5 = 10
2 x 4 = 8
2 x 3 = 6
2 x 2 = 4
2 x 1 = 2
```

```
for i in range(1,11):
    if (i % 2) == 0:
        result = " Even Number"
    else:
        result = " Odd Number"
    print(f"{i}: {result}")
```

```
1:  Odd Number
2:  Even Number
3:  Odd Number
4:  Even Number
5:  Odd Number
6:  Even Number
7:  Odd Number
8:  Even Number
9:  Odd Number
10: Even Number
```

```
for i in range(1,20):
    if (i % 5 == 0) and (i % 3 == 0):
        print(i,"Yes and No")
    elif (i % 5 == 0) or (i % 3 == 0):
        print(i, "Yes")
    else:
        print(i, "No")
```

```
1 No
2 No
3 Yes
4 No
5 Yes
6 Yes
7 No
8 No
9 Yes
10 Yes
11 No
12 Yes
13 No
14 No
15 Yes and No
16 No
17 No
18 Yes
19 No
```

```
lst = [5,3,6,8,9,2,7]
```

```
print("[", end="")
for i in lst:
    print(i, end = ", ")
print("\b\b]", end="")
```

```
[5, 3, 6, 8, 9, 2, 7]
```

```
sum = 0
for i in lst:
    print(i)
    sum = sum + i
print("==")
print(sum)
```

```
5
3
6
8
9
2
7
==
40
```

```
sum = 0
for i in lst:
    print(i, end=" + ")
```

```
sum = sum + i
print("\b\b=", sum)
```

5 + 3 + 6 + 8 + 9 + 2 + 7 = 40

```
# Odd number Sum
sum = 0
for i in lst:
    if (i % 2) != 0:
        print(i)
        sum = sum + i
print("==")
print(sum)
```

5
3
9
7
==
24

```
# Even number Sum
sum = 0
for i in lst:
    if (i % 2) == 0:
        print(i)
        sum = sum + i
print("==")
print(sum)
```

6
8
2
==
16

lst = [5, 3, 6, 8, 7, 2, 9]

```
max = 0
for i in lst:
    if i > max:
        max = i
print(max)
```

9

len(lst)

7

```
min = lst[0]
for i in lst:
    if i < min:
        min = i
print(min)
```

2

```
count = 0
a = 0
for i in lst:
    count = count + 1
print(count)
```

7

```
sum = 0
count = 0
for i in lst:
    sum = sum + i
    count = count + 1
print(sum / count)
```

5.714285714285714

```
for i in lst:
    if i % 2 == 0:
        print(i)
```

```
6
8
2
```

```
lst1 = []
```

```
for i in lst:
    if i % 2 == 0:
        lst1.append(i)
print(lst1)
```

```
[6, 8, 2]
```

```
lst2 = [] ; lst0 = []
```

```
for i in lst:
    if i % 2 == 0:
        lst2.append(i)
    else:
        lst0.append(i)
print(lst2)
print(lst0)
```

```
[6, 8, 2]
[5, 3, 7, 9]
```

✓ Unique Code

```
Lst = [4, 3, 6, 2, 7, 3, 5, 2]
```

```
LstUnique = []
```

```
for i in Lst:
    if i not in LstUnique:
        LstUnique.append(i)
print(LstUnique)
```

✓ Pattern

```
print(4 * "#")
```

```
####
```

```
for i in range(4):
    print(4 * "#")
```

```
####
####
####
####
```

```
for i in range(1,6):
    print(i * "#")
```

```
#
##
###
####
#####
```

```
for i in range(5,0,-1):
    print(i * "#")
```

```
#####
####
###
```

```
##  
#
```

```
for i in range(1,6):  
    print(i * "#")  
for i in range(5,0,-1):  
    print(i * "#")
```

```
#  
##  
###  
####  
#####  
#####  
#####  
###  
##  
#
```

```
for i in range(1,6):  
    print(i * " #")
```

```
#  
# #  
# # #  
# # # #  
# # # # #
```

```
n = 6  
for i in range(1,n,1):  
    print((n-i) * " " + "#")
```

```
    #  
    #  
  #  
  #  
#
```

```
n = 6  
for i in range(1,n,1):  
    print((n-i) * " " + i * "#")
```

```
    #  
    ##  
    ###  
    ####  
    #####
```

```
n = 6  
for i in range(1,n,1):  
    print(i * " " + (n-i) * "#")
```

```
#####  
####  
###  
##  
#
```

▼ Pyramid

```
n = 6  
for i in range(1, n, 1):  
    print((n-i) * " " + i * "#")  
    print(i * "#")
```

```
    #  
  #  
##  
###  
####  
#####  
#####  
#####
```

```
n = 6  
for i in range(1, n, 1):
```

```
print((n-i) * " " + i * "#", end="")
print(i * "#")
```

```
##
####
#####
#####
#####
#####
```

```
n = 6
for i in range(1, n, 1):
    print((n-i) * " " + i * "#", end="\b")
    print(i * "#")
```

```
#
###
####
#####
#####
#####
```

```
n = 6
for i in range(n, 0, -1):
    print((n-i) * " " + i * "#", end="\b")
    print(i * "#")
```

```
#####
#####
#####
#####
###
#
```

```
n = 6
for i in range(1, n, 1):
    print((n-i) * " " + i * "#", end="\b")
    print(i * "#")
for i in range(n, 0, -1):
    print((n-i) * " " + i * "#", end="\b")
    print(i * "#")
```

```
#
###
####
#####
#####
#####
#####
#####
#####
#####
###
#
```

Start coding or [generate](#) with AI.

✓ Examples

✓ Palin Droom

```
txt = 'python'
print(txt[::-1])
```

nohtyp

```
txt = ['python', 'civic', 'mongodb', 'radar']
```

```
for i in txt:
    print(i)
```

python
civic
mongodb
radar

```
for i in txt:
    if i == i[::-1]:
        print(i)
```

civic
radar

```
for i in txt:
    if i == i[::-1]:
        res = "\b"
    else:
        res = "NOT"
    print(f"{i} is {res} a Palindrome")
```

python is NOT a Palindrome
civic is a Palindrome
mongodb is NOT a Palindrome
radar is a Palindrome

```
txt = ['python', 'civic', 'mongodb', 'radar']
res=[]
for i in txt:
    if i == i[::-1]:
        res.append(i)
print(res)
```

['civic', 'radar']

```
x = [2, 4, 6, 7, 8]
fnd = int(input("Enter a Number: "))
for i in x:
    if i == fnd:
        output = "Found"
        break
    else:
        output = "Not Found"
print(fnd, output)
```

Enter a Number: 4
4 Found

```
print("i j")
for i in range(1,6,1):
    for j in range(2,4,1):
        print(i,j)
    print("-----")
```

i j
1 2
1 3

2 2
2 3


```
-----  
3 2  
3 3  
-----  
4 2  
4 3  
-----  
5 2  
5 3  
-----
```

```
for m in range(1,6,1):  
    for t in range(2,4,1):  
        print(t, "x", m, "=", t*m, end = "\t")  
    print("")
```

2 x 1 = 2	3 x 1 = 3
2 x 2 = 4	3 x 2 = 6
2 x 3 = 6	3 x 3 = 9
2 x 4 = 8	3 x 4 = 12
2 x 5 = 10	3 x 5 = 15

```
Lst = [10, 9.5, 10, 8, 7.5, 5, 10, 10]  
i = 1  
Rating = Lst[0]  
while (Rating >= 6):  
    print(Rating)  
    Rating = Lst[i]  
    i = i + 1
```

```
10  
9.5  
10  
8  
7.5
```

Start coding or [generate](#) with AI.